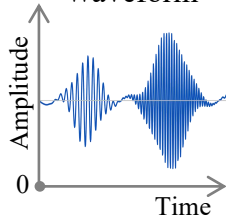


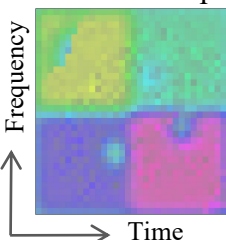
INPUT

Signal / TF patch
or SPD-field sample

Waveform



Controlled
SPD-field sample *



* hue encodes local
orientation and
brightness /
saturation reflects
anisotropy strength.

REPRESENTATION / PRE-PROCESSING

Representation

STFT and
log-amplitude map

Local gradients

Structure tensor
smoothing

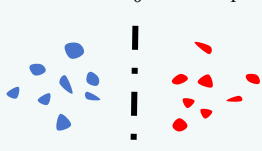
SPD regularization
 $\bar{S}(x) = J(x) + \lambda I_2$

Local SPD structure
field $\{S(x)\}$

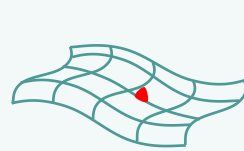
A) TRAINING / REFERENCE ESTIMATION

Reference modeling

Training samples
from H_0 and H_1



Pointwise Karcher
mean under AIRM



Class-conditional
reference fields

$$\bar{S}_0(x), \bar{S}_1(x)$$

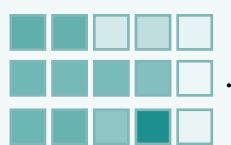
Discriminative
weight map $W(x)$

from between-class
separation &
within-class
dispersion

B) TESTING / LOCAL EVIDENCE

Local comparison

Test sample
structure field



AIRM distances to
references
 $d_0(x), d_1(x)$

Local distance-
difference evidence
 $\Delta(x) = d_0^2(x) - d_1^2(x)$

C) CALIBRATION

Threshold calibration

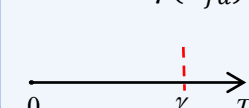
Independent H_0
calibration set



Sample-level statistics
on calibration data



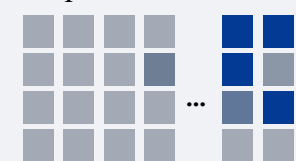
Threshold
calibration $\gamma(P_{fa})$



D) AGGREGATION / DECISION

Robust aggregation

Block-wise robust pooling
weighted block scores;
top-block selection



Sample-level statistic
 $T = (1 - \alpha)\mu_{all} + \alpha\mu_{top}$

Fixed- P_{fa} decision
 $T > \gamma(P_{fa}) \Rightarrow H_1$
; otherwise H_0

→ data / processing flow

---→ weighting influence